

TEACH YOURSELF LECTURE

Lecture 05

Application and Impacts of Computing

Chapter lecture — 17 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

Computer Science XII · Bilingual support: English + اردو

کمپیوٹر سائنس — بارہویں جماعت: خود سیکھیں، خود پرکھیں

17 teach-yourself lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XII

How to use this lecture

Learn this chapter **on your own**. Each topic is one A4 page. Read *Learn it*, try the worked example on paper, then answer *Check yourself* before looking at the answers. The chapter pack ends with a 75-mark exam. The green box repeats the idea in Urdu. ★ Golden topics are high-yield in board exams.

یہ لیکچر خود سیکھیں۔ مثال کاغذ پر حل کریں، پھر جواب دیکھنے سے پہلے خود کو پرکھیں۔

Lectures in this pack

Chapter 5: Application and Impacts of Computing. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 5.1 Machine Learning, Neural Networks and Deep Learning

L02. 5.1.1 Machine Learning (ML)

L03. 5.1.2 Neural Networks (NN)

L04. 5.1.3 Deep Learning (DL)

L05. 5.1.4 Components of Neural Networks

L06. 5.1.5 Components of Deep Learning

L07. Future of AI, NN and DL in Pakistan

L08. 5.2 Applications of Neural Networks and Deep Learning

L09. 5.3.1 Secure Collaboration (Authentication, MFA, Access Control)

L10. 5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

L11. 5.4.1 Common Security Threats (Malware, Phishing, DoS)

L12. 5.4.2 Identifying Security Threats

L13. 5.4.3 Apply Mitigation Techniques

L14. 5.5 Equity and Equal Access in Digital Collaboration

L15. 5.5.1 Functions of Equity and Equal Access

L16. 5.5.2 Collaboration Tools

L17. Career Opportunities

Application and Impacts of Computing

This chapter shows you how computers learn from data with machine learning, neural networks and deep learning, how to collaborate online without leaking secrets, how to protect data and stop cyber threats, and how to keep digital teamwork fair for every student in Sindh.

اس باب میں آپ سیکھیں گے کہ مشین لرننگ، نیورل نیٹ ورکس اور ڈیپ لرننگ کمپیوٹر کو ڈیٹا سے کیسے سکھاتے ہیں، ان لائن کام کو محفوظ کیسے رکھا جائے، ڈیٹا کی حفاظت اور سائبر خطرات کا مقابلہ کیسے کیا جائے، اور ڈیجیٹل تعاون کو سب کے لیے منصفانہ کیسے بنایا جائے۔

What you will learn

- Define **machine learning (ML)** and describe the three steps: data input, learning, prediction.
- Define **deep learning (DL)** and draw the relationship **AI $\supset$ ML $\supset$ DL**.
- List the six components of deep learning and give applications of NN and DL in daily life and in Pakistan.
- Protect data using encryption (including a Caesar cipher), strong passwords, the 3-2-1 backup rule and a firewall.
- Distinguish **equal access** from **equity** and describe the functions of equity in digital collaboration.
- Explain a **neural network (NN)** as a brain-inspired model with an input layer, hidden layers and an output layer.
- Name the eight components of a neural network and compute a neuron's output from given numbers.
- Explain **secure collaboration**: authentication, multi-factor authentication (MFA) and access control.
- Identify common threats (malware, phishing, DoS) and apply mitigation techniques.
- Name common collaboration tools and use them fairly and safely.

5.1 Machine Learning, Neural Networks and Deep Learning

LEARN IT

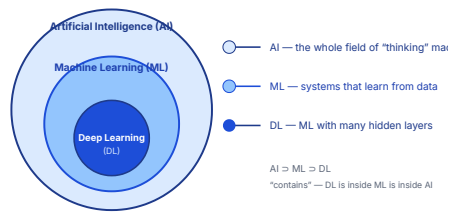
Machine learning (ML), **neural networks (NN)** and **deep learning (DL)** are important areas of **artificial intelligence (AI)**. **AI** is the broad field in which computers perform tasks that normally need human thinking, such as recognising a face, understanding speech or making a decision.

These technologies help computers **learn from data** and make useful decisions **without being explicitly programmed for every single step**. You do not write a separate rule for every possible face or every Urdu accent. You show the computer many examples, and it finds the patterns itself.

They are used in many modern systems:

- **Face unlock** on smartphones.
- **Speech recognition** (for example Siri, Alexa, or Urdu voice typing on a phone).
- **Photo search** categorisation (typing "car" in a gallery and seeing car photos).
- **Medical diagnosis** from X-rays and scans.
- **Language translation** (English to Urdu or Sindhi).

Remember the nesting: **AI is the biggest set**. Inside AI sits **ML**. Inside ML sits **DL**. Every deep-learning system is a kind of machine learning, and every machine-learning system is a kind of AI — but not every AI system is deep learning.



The set relationship $AI \supset ML \supset DL$. Deep learning is a method of machine learning; machine learning is a part of artificial intelligence.

WORKED EXAMPLE

Worked example: place each system in the correct set. Sara in Karachi uses three features on her phone.

1. **A chess app that follows fixed if-then rules** written by a programmer ("if the queen is attacked, move it"). This is **AI** (the machine decides), but it is **not ML**, because it does not learn from data.
2. **A JazzCash fraud filter** that was trained on thousands of past transactions and then flags a new unusual payment. This is **ML** (and therefore also AI). If it uses only a simple model with few layers, it may **not** be DL.
3. **Face unlock** that studied millions of face images through many hidden layers. This is **DL** — and therefore also ML and AI.

Exam sentence you can write: *All DL is ML, all ML is AI, but not all AI is ML, and not all ML is DL.*

REMEMBER IT

Russian-doll memory trick. Picture three boxes. The outer box is **AI**. Inside it is a smaller box **ML**. Inside that is the smallest box **DL**. You cannot take DL out of ML, and you cannot take ML out of AI.

COMMON MISTAKE

Mistake: "AI, ML and DL are three different, separate subjects."

Fix: They nest. $DL \subset ML \subset AI$. A deep-learning face-unlock system is all three at once.

اردو میں سمجھیں

مصنوعی ذہانت (AI) سب سے بڑا میدان ہے۔ اس کے اندر مشین لرننگ (ML) ہے، اور مشین لرننگ کے اندر ڈیپ لرننگ (DL) ہے۔ یہ ٹیکنالوجیز کمپیوٹر کو ڈیٹا سے سکھاتی ہیں تاکہ ہر قدم الگ الگ پروگرام کیے بغیر فیصلے کیے جا سکیں، جیسے چہرے سے فون کھولنا یا آواز پہچاننا۔

CHECK YOURSELF

1. Which is the largest set: AI, ML or DL?
2. Is every neural-network face-unlock system an example of AI? Why?
3. Give one modern system from this section that uses these technologies.

Answers: 1. AI ($AI \supset ML \supset DL$). 2. Yes — DL is inside ML, and ML is inside AI. 3. Any of: face unlock, speech recognition, photo search, medical diagnosis, language translation.

5.1.1 Machine Learning (ML)

LEARN IT

A **machine learning model** is the system that **learns from data and gives an output**. Think of it as the **trained brain** of a computer program.

In ordinary programming you write every rule yourself: *if marks ≥ 40 then "Pass"*. In machine learning you **do not list every rule**. You give the model many examples (data), it finds the patterns, and later it decides about new cases it has never seen.

How it works — three steps:

1. Data input → 2. Learning (find patterns) → 3. Prediction on new data

1. **Data input.** The model is given massive amounts of data — for example thousands of past BISE marksheets, or millions of crop photos from Sindh fields.
2. **Learning.** It studies the data and **learns patterns and rules automatically**. It might notice that students who scored low in Chapter 3 also failed the board exam, or that yellow spots on cotton leaves often mean a pest.
3. **Prediction.** After training, it can make **accurate predictions or decisions** when given **new, unseen data** — a new student's record, or a new leaf photograph.

Note from the source: there are many types of machine-learning models. A **neural network** is one important type. **Deep learning** is a highly advanced method of machine learning.

WORKED EXAMPLE

Worked example: a wheat-yield model for a farm near Sukkur.

Step 1 — Data input. For 8 years the farm recorded: rainfall (mm), temperature, fertiliser bags used, and the actual yield (maunds per acre).

Step 2 — Learning. The model studies those rows. It finds, for example, that yield rises with fertiliser up to a point, and falls when temperature is very high in April.

Step 3 — Prediction. This year the farmer types in this season's rainfall and temperature. The model has never seen this exact year before, but it predicts "about 32 maunds per acre." The farmer plans storage and a buyer in advance.

Nobody wrote a giant list of "if rain = 120 and temp = 38 then yield = ...". The model learned the pattern from data.

REMEMBER IT

D-L-P. Machine learning is **Data in, Learn patterns, Predict**. "Data Learns to Predict."

COMMON MISTAKE

Mistake: "The computer is intelligent on day one."

Fix: An ML model is useful only **after training** on enough good data. Garbage data → garbage predictions. Also, neural networks are *one* type of ML, not the only type; deep learning is a *method* that uses deep neural networks.

اردو میں سمجھیں

مشین لرننگ ماڈل وہ نظام ہے جو ڈیٹا سے سیکھ کر نتیجہ دیتا ہے — کمپیوٹر پروگرام کا تربیت یافتہ دماغ۔ پہلے بہت سا ڈیٹا دیا جاتا ہے، پھر ماڈل خود اصول اور پیٹرن سیکھتا ہے، اور تربیت کے بعد نئے ان دیکھے ڈیٹا پر پیش گوئی کرتا ہے۔ نیورل نیٹ ورک مشین لرننگ کی ایک اہم قسم ہے، اور ڈیپ لرننگ اس کا ایک اعلیٰ طریقہ ہے۔

CHECK YOURSELF

1. What are the three steps of how an ML model works?
2. What is a machine-learning model compared to in the source?
3. After training, what kind of data can the model handle?

Answers: 1. Data input, learning, prediction. 2. The trained brain of a computer program. 3. New, unseen data.

5.1.2 Neural Networks (NN) ★ Golden

LEARN IT

A **neural network** is a computer model that **learns patterns from data**. It is **directly inspired by the human brain**, but it works as a **mathematical computer system** — not as living tissue.

Your brain has billions of nerve cells (**neurons**) that pass signals to each other. A neural network copies this idea with small processing units also called **neurons** or **nodes**. These nodes are **connected** and **pass information from one layer to another**, performing calculations at each step.

A typical neural network has three kinds of layer:

- **Input layer** — receives the raw data (pixels of a face, samples of a sound, marks of a student).
- **Hidden layer(s)** — the “thinking” in the middle. Nodes here combine the inputs and learn features (eye distance, nose shape, an edge in a photo).
- **Output layer** — gives the final answer (unlock / keep locked; “this is a cat”; “pass / fail”).

Input layer → Hidden layer(s) → Output layer

WORKED EXAMPLE

Practical example from the source: face unlock.

Step 1 — Input layer. You look at your phone. The camera captures your face as numbers (brightness of each pixel). Those numbers enter the input layer.

Step 2 — Hidden layers. The system studies face features such as **eye distance** and **nose shape**. Early hidden nodes may detect edges and shadows; later hidden nodes combine those into “this looks like Sara’s face.” It compares them with stored data from when you first set up the phone.

Step 3 — Output layer. If the features match, the phone **unlocks**. If they do not (your brother holds the phone), it stays locked.

The same three-layer story works for an NADRA-style biometric gate at a college: camera (input) → feature matching (hidden) → open the turnstile or not (output).

REMEMBER IT

I-H-O. Input receives, Hidden thinks, Output decides. “In Hidden Out.” A neural network is a **brain-inspired maths machine**, not a copy of a living brain.

COMMON MISTAKE

Mistake: “A neural network is an actual human brain inside the computer.”

Fix: It is only **inspired by** the brain. It is mathematics: numbers, multiplications and additions running on ordinary hardware.

اردو میں سمجھیں

نیورل نیٹ ورک ایک کمپیوٹر ماڈل ہے جو ڈیٹا سے پیٹرن سیکھتا ہے۔ یہ انسانی دماغ سے متاثر ہے مگر ایک ریاضیاتی نظام کے طور پر کام کرتا ہے۔ چھوٹی اکائیاں نیورون یا نوڈ کہلاتی ہیں، جو تھوں میں جڑی ہوتی ہیں: ان پٹ تھ ڈیٹا لیتی ہے، پوشیدہ تھ سوچتی ہے، اور آؤٹ پٹ تھ فیصلہ دیتی ہے — جیسے چہرے سے فون کھولنا۔

EXAM TIP

Very common questions: “What is a neural network?” (2–3 marks), “Name its three layers and explain each with the face-unlock example” (5 marks), and “How is a neural network related to the human brain?”

Full-mark recipe: (1) definition — computer model that learns patterns from data, inspired by the brain, mathematical; (2) neurons/nodes connected across layers; (3) Input / Hidden / Output, each in one sentence; (4) walk the camera → features (eye distance, nose shape) → unlock path. Draw the three boxes if the paper asks for a diagram.

CHECK YOURSELF

1. What are the small processing units of a neural network called?
2. In face unlock, which layer studies eye distance and nose shape?
3. Is a neural network a living brain? Explain in one sentence.

Answers: 1. Neurons or nodes. 2. Hidden layers. 3. No — it is inspired by the brain but works as a mathematical computer system.

5.1.3 Deep Learning (DL)

LEARN IT

Deep learning is a machine-learning method in which computers use **multi-layer neural networks** to learn **complex patterns** from data.

The word **“deep”** is used because data passes through **several hidden layers** before the system produces the final output. Each layer learns a **different level of detail**:

- **Early layers** learn simple features — edges, shapes, colours.
- **Deeper layers** combine those simple features to understand complex patterns — a whole face, or a specific car model (a Cultus versus a Corolla).

A network with only one or two hidden layers is a **shallow** neural network. Deep learning is always “deep”: it needs **huge amounts of data** and **processing power**.

Neural network vs deep learning

	Neural network (the model)	Deep learning (the method)
What it is	A system/model that has layers, nodes and connections to help it learn.	A specific method that trains neural networks containing many hidden layers .
Depth	Can be simple (shallow) with just one or two hidden layers.	Always “deep,” requiring huge amounts of data and processing power to learn complex patterns.
Role	The architecture (the “machine”).	The way we train a deep version of that machine.

WORKED EXAMPLE

Real-world example from the source: photo gallery search.

A modern gallery app on a smartphone uses deep learning to recognise faces, objects or places in thousands of photos.

Step 1. During training (done by the company, using millions of images) early layers learned edges and colours; deeper layers learned “wheel”, “petal”, “minaret”.

Step 2. The trained network has already scanned your phone’s photos and tagged them.

Step 3. You type “car” or “flower” in the search bar. The app instantly shows related photos because its deep neural network has already learned visual patterns from millions of training images.

A student in Hyderabad can type “Ajrak” or “cricket” and jump to those pictures without scrolling for ten minutes.

REMEMBER IT

Deep = many hidden layers. Shallow NN: 1–2 hidden layers. Deep NN: many hidden layers, huge data, strong computers. Simple features first, complex meaning later.

COMMON MISTAKE

Mistake: “Deep learning and neural networks are two names for the same thing.”

Fix: A neural network is the **model** (layers + nodes + connections). Deep learning is the **method** of training networks that have **many** hidden layers. A small NN with one hidden layer is not deep learning.

اردو میں سمجھیں

ڈیپ لرننگ مشین لرننگ کا وہ طریقہ ہے جس میں کئی پوشیدہ تہوں والے نیورل نیٹ ورک استعمال ہوتے ہیں۔ “ڈیپ” اس لیے کہ ڈیٹا کئی تہوں سے گزر کر پیچیدہ پیٹرن سیکھتا ہے: پہلی تہیں کنارے اور رنگ سیکھتی ہیں، گہری تہیں پورا چہرہ یا گاڑی کی قسم پہچانتی ہیں۔ یہ طریقہ بہت سے ڈیٹا اور طاقتور کمپیوٹنگ کا طالب ہے۔

CHECK YOURSELF

1. Why is the method called “deep” learning?
2. What do early layers learn, and what do deeper layers learn?
3. Can a neural network with one hidden layer be called deep learning?

Answers: 1. Because data passes through several hidden layers. 2. Early: simple features (edges, shapes, colours). Deeper: complex patterns (a whole face, a car model). 3. No — that is a shallow network; DL always uses many hidden layers.

5.1.4 Components of Neural Networks ★ Golden

LEARN IT

The components of a neural network can be understood through its **layered architecture**. Data **enters**, gets **processed**, and **exits**, with mathematical adjustments happening inside.

#	Component	Description
1	Input layer	Receives the initial raw data (for example image pixels, sound waves, text).
2	Neurons / nodes	Small processing units that receive, calculate, and pass information forward.
3	Hidden layer(s)	The layers between input and output. They process the data and learn patterns from it.
4	Weights	Numbers that show the importance or strength of connections between neurons.
5	Bias	An extra constant value added to help the model adjust its result and improve accuracy.
6	Activation function	A mathematical gate that decides whether the information is important enough to move to the next layer.
7	Output layer	Gives the final computed result, decision or prediction (for example "This is a cat").
8	Loss function	Checks the error (how much the model's answer is wrong compared with reality) so the network can correct itself.

What happens inside one neuron

Each neuron does a short calculation:

- Multiply each input x_i by its weight w_i .
- Add those products together, then add the **bias** b . This total is the **weighted sum** s .
- Pass s through an **activation function**. A simple exam-friendly gate is a **threshold (step) function**: if s is at least the threshold, the neuron **fires** (output = 1); otherwise output = 0.

In symbols: $s = (x1 \times w1) + (x2 \times w2) + \dots + b$, then $y = 1$ if $s \geq \text{threshold}$, else $y = 0$.



One neuron: inputs $\times$ weights are summed, bias is added, then an activation function decides the output y .



A small neural network: 3 input nodes, 4 hidden nodes, 2 output nodes. Each line is a weighted connection.

WORKED EXAMPLE

Worked example: neuron calculation with numbers. A college in Larkana uses a tiny network to guess whether a student is ready for a test. One hidden neuron receives two inputs:

Symbol	Meaning	Value
$x1$	Hours studied yesterday (scaled 0 to 1)	0.8
$x2$	Attendance this month (scaled 0 to 1)	0.5
$w1$	Weight of hours studied	0.4
$w2$	Weight of attendance	0.6
b	Bias	0.1
threshold	Activation gate	0.5

Step 1 — weighted products.

$x1 \times w1 = 0.8 \times 0.4 = \mathbf{0.32}$

$x2 \times w2 = 0.5 \times 0.6 = \mathbf{0.30}$

Step 2 — add bias to get the weighted sum s .

$s = 0.32 + 0.30 + 0.1 = \mathbf{0.72}$

Step 3 — activation (threshold 0.5). Is $0.72 \geq 0.5$? **Yes**, so the neuron fires: $y = 1$ (ready for the test).

Step 4 — a second student with $x1 = 0.2$ (almost no study) and $x2 = 0.3$:

$s = (0.2 \times 0.4) + (0.3 \times 0.6) + 0.1 = 0.08 + 0.18 + 0.1 = \mathbf{0.36}$

0.36 is **less than** 0.5, so $y = \mathbf{0}$ (not ready). The same neuron, different inputs, different decision.

You can check the first student in Python 3:

```
x1, x2 = 0.8, 0.5
w1, w2 = 0.4, 0.6
b = 0.1
s = x1 * w1 + x2 * w2 + b
y = 1 if s >= 0.5 else 0
print(round(s, 2))
print(y)
```

```
OUTPUT
0.72
1
```

Loss function, briefly. If the true answer for the first student was actually "not ready" (true $y = 0$) but the model said 1, the loss is high. Training then slightly changes weights and bias so the next guess is closer to reality.

REMEMBER IT

Weights = importance. **Bias = extra nudge.** **Activation = yes/no gate.** **Loss = how wrong.** A large weight means that input matters more. Bias shifts the sum up or down so the gate can open at the right place.

COMMON MISTAKE

Mistake: Forgetting to add the bias, or multiplying instead of adding after the products.

Fix: Always **multiply first** (each x by its w), **then add** the products, **then add b** , **then** apply the activation. Write the four lines in the exam as we did above.

آرڈو میں سمجھیں

نیورل نیٹ ورک کے اجزاء ہیں: ان پٹ تہ، نیورون، پوزہید تہ، وزن (Weights)، بانیس (Bias)، ایکٹیویشن فنکشن، آؤٹ پٹ تہ اور لاس فنکشن۔ ہر نیورون ان پٹ کو وزن سے ضرب دے کر جمع کرتا ہے، بانیس جوڑتا ہے، پھر ایک گیٹ فیصلہ کرتا ہے کہ سگنل آگے جانے یا نہیں۔ وزن تعلق کی طاقت بتاتا ہے، لاس فنکشن غلطی بتاتا ہے تاکہ ماڈل خود درست ہو سکے۔

EXAM TIP

Golden questions: "List the components of a neural network" (often 8 names + one line each), "What are weights and bias?", and a **numerical neuron**: given x , w , b and a threshold, find s and y .

Show every arithmetic step. Underline the final y . If asked for a diagram, draw Input $\rightarrow$ Hidden $\rightarrow$ Output with circles and connecting lines, and label weights on the lines. Mention the loss function as the component that measures error so the network can correct itself.

CHECK YOURSELF

- What do weights show?
- Using $x1 = 1$, $x2 = 0$, $w1 = 0.5$, $w2 = 0.5$, $b = 0$, threshold = 0.4, what is y ?
- What does the loss function check?

Answers: 1. The importance or strength of connections between neurons. 2. $s = 1 \times 0.5 + 0 \times 0.5 + 0 = 0.5$; $0.5 \geq 0.4$ so $y = 1$. 3. The error — how much the model's answer is wrong compared with reality.

5.1.5 Components of Deep Learning

LEARN IT

Deep learning relies on specific advanced components to achieve its high accuracy. These components **differentiate it from basic machine learning**.

#	Component	What it means
1	Large input data	Deep learning is "data hungry." It needs massive datasets (thousands or millions of records) for better learning and accuracy.
2	Many hidden layers	Uses dozens or hundreds of hidden layers to learn deeper, more intricate patterns.
3	Automatic feature learning	Unlike older models where humans had to manually highlight important features, deep learning extracts important features by itself .
4	Learning algorithm	Math routines (like backpropagation) that help the system reduce mistakes and improve learning continuously.
5	High computing power	Processing millions of data points across deep layers needs specialised, strong hardware, such as GPUs (Graphics Processing Units).
6	Training process	The model is trained again and again in loops called epochs to gradually improve its final performance.

A **GPU** was originally made for video games' graphics, but it can do thousands of small multiplications in parallel — exactly what deep networks need. An **epoch** is one full pass through the training data. Ten epochs means the model has seen every training example ten times, adjusting weights a little each time.

Backpropagation means sending the error (from the loss function) **backwards** through the layers so each weight can be nudged in the direction that reduces the next error.

WORKED EXAMPLE

Worked example: cotton-leaf disease detector for a farm in Mirpurkhas. Match each DL component.

1. **Large input data.** Researchers collect 80,000 labelled photos: healthy leaves and leaves with whitefly damage.
2. **Many hidden layers.** The network has 20 hidden layers so it can go from "green edge" to "this exact pest pattern."
3. **Automatic feature learning.** No agronomist types in "look for yellow circles." The network finds the useful spots itself.
4. **Learning algorithm.** After a wrong guess ("healthy" when it was diseased), backpropagation lowers the weights that caused the mistake.
5. **High computing power.** Training runs on a GPU workstation at a university lab in Karachi; a school laptop would take weeks.
6. **Training process.** They train for 30 epochs. After epoch 1 accuracy is 62%; after epoch 30 it is 94% on photos it has not trained on.

REMEMBER IT

D-L-A-L-G-T. Data (lots), Layers (many), Automatic features, Learning algorithm, GPU power, Training epochs. "Deep Learners Always Love Good Training."

COMMON MISTAKE

Mistake: "Deep learning is just 'a neural network'."

Fix: Name the six extras: huge data, many hidden layers, automatic features, a learning algorithm (backpropagation), GPUs, and repeated epochs. A shallow NN with a small spreadsheet is ordinary ML, not DL.

اردو میں سمجھیں

ڈیپ لرننگ کو عام مشین لرننگ سے یہ چیزیں الگ کرتی ہیں: بہت بڑا ڈیٹا، بہت سی پوشیدہ تہیں، خود بخود خصوصیات نکالنا، سیکھنے کا الگورتھم جیسے بیک پروپیگیشن (Backpropagation)، طاقتور ہارڈویئر جیسا کہ جی پی یو (GPU)، اور بار بار تربیت جسے ایپوک (Epoch) کہتے ہیں۔

CHECK YOURSELF

1. Why is deep learning called "data hungry"?
2. What is an epoch?
3. Name the hardware often used for DL, and the learning algorithm named in the source.

Answers: 1. It needs massive datasets (thousands or millions of records) for accuracy. 2. One training loop — one full pass through the training data. 3. GPUs (Graphics Processing Units); backpropagation.

Future of AI, NN and DL in Pakistan

LEARN IT

Machine learning, neural networks and deep learning will play a key role in Pakistan's future. The **National AI Policy 2025** highlights how artificial intelligence is now part of national planning, focusing on **innovation, skill-building, ethical use and digital growth**.

Sector	Potential impact in Pakistan
Education	Smart learning systems, online tutoring, automatic assessments, and finding where students need specific help.
Healthcare	Disease detection, medical image analysis, and managing hospital data. Crucial for rural areas lacking specialist doctors.
Agriculture	Monitoring crops, predicting weather, managing irrigation, and estimating yields — highly important for Pakistan's economy.
Business and banking	Fraud detection, customer support (chatbots), market forecasting, and intelligent risk analysis.
Local languages	Speech recognition, translation and text-to-speech tools for Urdu, Sindhi and other regional languages.

To benefit from this future, Pakistan needs skilled individuals. You should focus on **programming (such as Python), mathematics, data analysis, problem-solving** and the **ethical use of technology**.

WORKED EXAMPLE

Worked example: pick a sector and write a three-line "impact paragraph" for 5 marks.

Agriculture: A cotton grower in Sanghar photographs leaves on a cheap phone. A deep-learning model trained on Pakistani crop images flags whitefly early. The farmer sprays only the affected block, saves pesticide cost, and protects yield — which matters because agriculture is a large part of Pakistan's economy.

REMEMBER IT

Five sectors: E-H-A-B-L. Education, Healthcare, Agriculture, Business/banking, Local languages. Policy words: innovation, skills, ethics, digital growth.

COMMON MISTAKE

Mistake: Writing only "AI will make Pakistan modern" with no sector or example.

Fix: Name the National AI Policy 2025, pick two sectors from the table, and give a local example (rural hospital without a specialist; Urdu/Sindhi speech tools; bank fraud).

اردو میں سمجھیں

قومی اے آئی پالیسی 2025 کے مطابق مصنوعی ذہانت پاکستان کی منصوبہ بندی کا حصہ ہے: جدت، مہارت، اخلاقی استعمال اور ڈیجیٹل ترقی۔ تعلیم، صحت، زراعت، بینکنگ اور اردو و سندھی جیسی مقامی زبانوں میں اس کے اثرات ہوں گے۔ طلبہ کو پائتھن، ریاضی، ڈیٹا کا تجزیہ، مسئلہ حل کرنا اور ٹیکنالوجی کا اخلاقی استعمال سیکھنا چاہیے۔

CHECK YOURSELF

1. Name the national policy mentioned for AI in Pakistan.
2. Why is AI in healthcare especially important for rural areas?
3. List three skills students should learn to prepare for this future.

Answers: 1. National AI Policy 2025. 2. Those areas often lack specialist doctors; AI can help with disease detection and medical image analysis. 3. Any three of: programming (Python), mathematics, data analysis, problem-solving, ethical use of technology.

5.2 Applications of Neural Networks and Deep Learning

LEARN IT

Neural networks and deep learning are widely used in modern life. Many intelligent systems use them in the background to **recognise information, make predictions and support users**.

#	Application	What it does	Example
1	Image recognition	Identifies objects, faces or patterns in images. Used in security cameras, photo search and biometric locks.	Unlocking a smartphone using face recognition; a camera at a Karachi plaza matching a face with a staff list.
2	Speech recognition	Converts spoken words into digital text or commands. Deep learning helps with accents, pronunciation and voice patterns.	A student speaks to Google Assistant or Alexa to set a morning alarm; Urdu voice search on a phone.
3	Natural Language Processing (NLP)	Helps computers understand, process and respond to human language grammatically and in context. Core of chatbots, translators and AI writers.	Using Google Translate to turn a difficult English paragraph into Urdu; a bank chatbot answering in Urdu.
4	Healthcare systems	Analyse complex medical images (X-rays, CT scans, MRIs) to detect diseases early and accurately.	A hospital system highlighting abnormal patterns in a lung X-ray that might indicate pneumonia or tumours — useful at a DHQ hospital that has few radiologists.
5	Recommendation systems	Analyse past behaviour and preferences to suggest content, products or services.	A student watches computer-science videos on YouTube; the platform recommends more programming tutorials.
6	Autonomous vehicles	Self-driving cars use neural networks to process real-time camera and radar data: roads, signals, pedestrians, obstacles.	A self-driving car detects a pedestrian crossing and automatically applies the brakes.
7	Fraud detection	Banks compare current activity with previous spending patterns to catch unusual transactions.	An HBL or Meezan app flags a sudden Rs 80,000 payment from a new city at 3 a.m.
8	Assistive technologies	AI supports people with disabilities.	Speech-to-text tools for users who have difficulty typing; screen readers that speak Urdu text aloud.

WORKED EXAMPLE

Worked example: match the app to the application name — a day in Ahmed's life in Hyderabad.

- 7:00 am — phone unlocks by face. → **Image recognition**.
- 7:05 am — "Ok Google, set alarm for Maghrib." → **Speech recognition**.
- 8:00 am — Google Translate helps with an English article. → **NLP**.
- 4:00 pm — YouTube suggests a Python tutorial after he watched Chapter 5 notes. → **Recommendation system**.
- 9:00 pm — JazzCash texts "We blocked a payment that does not match your pattern." → **Fraud detection**.

REMEMBER IT

Eight applications: I-S-N-H-R-A-F-A. Image, Speech, NLP, Healthcare, Recommendations, Autonomous vehicles, Fraud, Assistive tech. "I See New Hospitals Recommend Autonomous Fraud-Assist."

COMMON MISTAKE

Mistake: Mixing up speech recognition and NLP.

Fix: Speech recognition turns **sound into text** (or a command). NLP understands the **meaning of language** (grammar, context, translation, chat replies). You can have NLP on typed text with no microphone at all.

اردو میں سمجھیں

نیورل نیٹ ورکس اور ڈیپ لرننگ کی مشہور ایپلی کیشنز یہ ہیں: تصویر کی پہچان، آواز کی پہچان، قدرتی زبان کی پروسیسنگ (NLP)، صحت کے نظام، تجویز کرنے والے نظام، خود چلنے والی گاڑیاں، فراڈ کی نشاندہی اور معاون ٹیکنالوجیز۔ روزمرہ میں چہرے سے فون کھولنا، یوٹیوب کی تجاویز اور بینک کا مشکوک لین دین پکڑنا سب اسی خاندان سے ہیں۔

CHECK YOURSELF

- Which application converts spoken words into text or commands?
- Give the source example of a recommendation system.
- How do self-driving cars use neural networks?

Answers: 1. Speech recognition. 2. YouTube recommending more programming tutorials after a student watches computer-science videos. 3. They process real-time camera and radar data to understand roads, traffic signals, pedestrians and obstacles (for example braking for a pedestrian).

5.3.1 Secure Collaboration (Authentication, MFA, Access Control) ★ Golden

LEARN IT
Digital tools allow people to **share files, attend online classes, edit documents and work together from different locations**. Collaboration is useful only when **data is protected** and **communication remains safe**.
Secure collaboration means using digital tools to work together in a **safe, controlled and responsible** way. It depends on several key elements.

i) Authentication / identity verification

Authentication is **confirming the identity of a user before allowing access**. Common methods:

- **Username and password**
- **PIN** (a short number code)
- **OTP — One-Time Password**, a code that works only once, usually sent by SMS or an authenticator app
- **Biometrics** — fingerprint or face

Source example: a **banking app sending an OTP to your phone before login**.

Multi-factor authentication (MFA) and two-factor authentication (2FA)

A **factor** is a category of proof:

- **Something you know** — password or PIN.
- **Something you have** — your phone, an OTP, a bank SMS.
- **Something you are** — fingerprint, face (biometrics).

Multi-factor authentication (MFA) means the system asks for **two or more different factors** before it trusts you. **Two-factor authentication (2FA)** is the common case of MFA that uses **exactly two** factors — usually password + OTP.

If a thief in another city steals only your password, MFA still blocks them because they do not have your phone (or your fingerprint).

ii) Access control

Access control decides **what an authenticated user is allowed to do**: View, Edit, Comment, Download, Share.

Authentication answers **“Who are you?”** Access control answers **“What may you do?”**

Source example: a teacher shares a Google Doc with students as **“view only”**, but gives group leaders **“edit”** permissions.

iii) Accountability and tracking

User actions are linked to **verified accounts**. You can see who opened, edited, deleted or shared a file. Example: a school **LMS** (Learning Management System) tracks which teacher uploaded an assignment and which student submitted it.

iv) Data privacy

Data privacy means protecting personal and sensitive information and using it **only for the purpose it was collected**. It involves:

- **Protection** — keeping names, numbers, marks and passwords secure.
- **Controlled sharing** — sharing only with the right people.
- **Confidentiality** — data stays private during storage and transmission.
- **Compliance** — following legal and ethical rules for data handling.

v) Safe communication / transmission

Exchanging information through digital platforms **securely**, protecting messages from unauthorised interception or wiretapping. Example: a teacher shares class instructions through an **official, encrypted LMS** rather than an open public forum or an unknown WhatsApp broadcast.

WORKED EXAMPLE
Worked example: a Class XII Google Classroom in Sukkur.
Step 1 — Authentication. Each student signs in with a school email and password. The college turns on MFA: after the password, Google sends an OTP to the student's registered Jazz/Zong number.
Step 2 — Access control. All 40 students can **view** the assignment sheet. Only the four group leaders can **edit** the shared report. Nobody except the teacher can **delete** the class materials. A student who left the college is removed from the class (access revoked).
Step 3 — Accountability. Version history shows that Ali edited paragraph 2 at 8:14 pm and that Hina submitted the PDF at 9:02 pm.
Step 4 — Privacy and safe transmission. Marks stay inside Classroom; the teacher does not post CNIC numbers or phone numbers in a public Facebook group. Traffic to Google is encrypted, unlike an open public forum.
If Ali shares his password with a cousin, authentication is broken. If the teacher gives every student “edit” on the marksheet, access control is broken even though login still works.

REMEMBER IT
A-A-A-P-S. Authentication, Access control, Accountability, Privacy, Safe transmission. MFA = at least two different factor types (know / have / are). 2FA is MFA with two factors.

COMMON MISTAKE
Mistake: Treating authentication and access control as the same thing.
Fix: Login only proves identity. After login, access control still limits View / Edit / Comment / Download / Share. A student who is authenticated must not automatically get teacher rights.

اردو میں سمجھیں
محفوظ تعاون کا مطلب ہے ڈیجیٹل ٹولز سے محفوظ، کنٹرول شدہ اور ذمہ دارانہ انداز میں ساتھ کام کرنا۔ تصدیق (Authentication) یہ جانچتی ہے کہ صارف کون ہے؛ رسائی کنٹرول (Access Control) یہ طے کرتا ہے کہ وہ کیا کر سکتا ہے (صرف دیکھنا، ترمیم، تبصرہ)۔ کثیر عنصری تصدیق (MFA) دو یا زیادہ مختلف ثبوت مانگتی ہے، جیسے پاس ورڈ اور او ٹی پی (OTP)۔ حساب دہی، ڈیٹا پرائیویسی اور محفوظ ترسیل بھی ضروری ہیں۔

EXAM TIP
Typical items: “Define secure collaboration”, “Differentiate authentication and access control”, “What is MFA/2FA?”, “Explain five elements of secure collaboration with examples.”
Write the five headings (authentication, access control, accountability, privacy, safe transmission). For MFA, name the three factor types and say that 2FA is password + OTP in a banking app. Use the Google Doc “view only vs edit” example — examiners look for it.

CHECK YOURSELF
1. What question does authentication answer, and what question does access control answer?
2. Password + OTP is an example of which idea?
3. Give the source example of access control in a Google Doc.

Answers: 1. Authentication: who are you? Access control: what may you do? 2. Two-factor / multi-factor authentication (2FA/MFA). 3. Teacher shares the document as view-only with students but gives group leaders edit permission.

5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

LEARN IT

Data protection means keeping digital information safe from **unauthorised access, misuse, loss, damage or accidental deletion**.

1. Data encryption

Encryption changes readable data (**plaintext**) into a coded form called **ciphertext**. Unauthorised users cannot understand ciphertext without the key.

- **Symmetric encryption** — the **same key** is used to encrypt and to decrypt.
- **Asymmetric encryption** — **two keys**: a **public key** to encrypt, a **private key** to decrypt.

A simple exam cipher is the **Caesar cipher** (a symmetric substitution cipher). You shift every letter a fixed number of places along the alphabet. Shift 3 is the classic key: A → D, B → E, ..., Z → C (the alphabet wraps around).

2. Strong passwords

A strong password includes **uppercase, lowercase, numbers and symbols**. It resists **brute-force** guessing (trying every possible combination). Source example: Zk@2026#Pass.

Weak: ahmed123, pakistan, sindh, your date of birth. Never reuse the same password on the college LMS and on JazzCash.

3. Backup and recovery — the 3-2-1 rule

A **backup** is a copy of data in a safe location (cloud or external drive) so you can **restore** it after hardware damage, ransomware or accidental deletion.

The **3-2-1 backup rule** is a practical standard:

- **3** copies of the data (the original plus two backups).
- **2** different types of storage media (for example laptop disk + USB, or disk + cloud).
- **1** copy **off-site** (a different building or the cloud), so a fire or theft at home does not destroy every copy.

4. Firewall and antivirus software

A **firewall** is a security **barrier** that **monitors network traffic** and **blocks suspicious internet access**. Think of it as a checkpoint on the road between your device (or college lab) and the internet.

Antivirus software detects and removes harmful software (viruses, spyware and other malware) from the **local machine**.

Firewall ≈ guards the **door to the network**. Antivirus ≈ searches **inside the house** for intruders already present.

WORKED EXAMPLE

Worked example A — Caesar cipher, shift = 3. Encrypt the plaintext **SINDH**, then decrypt the result.

Write the alphabet and count three letters forward. Wrap after Z back to A.

Plain	S	I	N	D	H
+3	V	L	Q	G	K

Ciphertext = **VLQ GK**.

Decrypt: shift each letter **back** by 3: V → S, L → I, Q → N, G → D, K → H. You recover **SINDH**.

Spaces and punctuation are often left unchanged. The same key (3) is used both ways, so Caesar is **symmetric**.

Second check: **CAT** + 3 → C→F, A→D, T→W → **FDW**.

```
def caesar(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord("A") if ch.isupper() else ord("a")
            result += chr((ord(ch) - base + shift) % 26 + base)
        else:
            result += ch
    return result

print(caesar("SINDH", 3))
print(caesar("VLQ GK", -3))
print(caesar("CAT", 3))
```

OUTPUT

VLQ GK

SINDH

FDW

Worked example B — 3-2-1 for Fatima's Class XII project in Hyderabad.

1. **Copy 1 (original)**: project folder on her laptop hard disk.
2. **Copy 2 (different media)**: the same folder on a USB drive she keeps in her bag.
3. **Copy 3 (off-site)**: uploaded to Google Drive (cloud servers are not in her house).

That is 3 copies, 2 media types (internal disk + USB, with cloud as the off-site copy), 1 off-site. When her brother deletes the laptop folder, she restores from USB or Drive.

Password check: Zk@2026#Pass has uppercase Z, lowercase letters, digits 2026, and symbols @ and #. fatima has none of those extras — reject it.



REMEMBER IT

Cipher = lock the message. Password = lock the door. 3-2-1 = spare keys in three places. Firewall = gate. Antivirus = indoor search. Caesar: shift letters; same key both ways (symmetric).

COMMON MISTAKE

Mistake 1: Shifting the wrong way, or forgetting wrap-around (X, Y, Z + 3 → A, B, C).

Mistake 2: Two copies on the same laptop (Documents and Desktop) and calling it 3-2-1. That is still one disk in one house.

Fix: Count letters on a written alphabet. For backups, different media and one copy that would survive a house fire.

اردو میں سمجھیں

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔ انکریپشن پڑھے جانے والے متن کو سائفر ٹیکسٹ بنا دیتی ہے؛ سیزر سائفر میں ہر حرف کو مقررہ جگہوں سے آگے سرکایا جاتا ہے۔ مضبوط پاس ورڈ میں بڑے، چھوٹے حروف، عدد اور علامات ہوں۔ 1-2-3 ایک آپ: تین کاپیاں، دو قسم کے میڈیا، ایک کاپی باہر (آف سائٹ)۔ فائر وال نیٹ ورک ٹریفک روکتی ہے؛ اینٹی وائرس مقامی نقصان دہ سافٹ ویئر بناتا ہے۔

CHECK YOURSELF

1. Encrypt PAK with a Caesar shift of 3.
2. What do the 3, the 2 and the 1 stand for in 3-2-1 backup?
3. In asymmetric encryption, which key encrypts and which key decrypts?

Answers: 1. SDN (P→S, A→D, K→N). 2. 3 copies, 2 different media, 1 off-site copy. 3. Public key encrypts; private key decrypts.

5.4.1 Common Security Threats (Malware, Phishing, DoS) ★ Golden

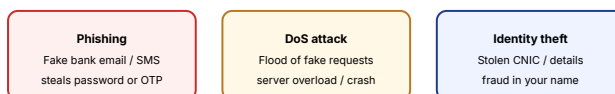
LEARN IT

A **security threat** is any action, event or weakness that can **harm a computer system or its data**. **Mitigation techniques** are protective measures used to prevent threats or reduce their impact. This section names the common threats; later sections show how to spot them and how to reduce the damage.

#	Threat	What it is	Example
1	Malware	Harmful software designed to damage a computer, steal data, or gain unauthorised access. Includes viruses, worms, spyware, ransomware and Trojans .	Installing cracked software that secretly corrupts files, or a "free game" USB that encrypts a college lab's coursework (ransomware).
2	Phishing	A social-engineering trick that deceives users into sharing personal information (passwords, bank details, OTPs) using fake emails or websites that look genuine.	An SMS that looks like JazzCash or HBL asking you to "verify" by typing your OTP on a fake page.
3	Unauthorised access	Entering a system, account or file without permission. Often caused by weak passwords, shared passwords, or leaving devices unlocked.	A classmate using an unlocked lab PC to open someone else's Google Classroom.
4	Data theft / identity theft	Data theft is stealing or copying data without permission. Identity theft is using another person's stolen details (name, CNIC, bank details) to pretend to be them online and commit fraud.	A leaked list of CNICs used to open a SIM or a wallet in someone else's name.
5	Denial of Service (DoS)	A website or system is flooded with thousands of fake requests, causing it to slow down or crash, so real users cannot access the service.	Hackers flooding an online exam portal so students cannot log in on BISE result day or during an e-test.
6	Human error	Mistakes by legitimate users: accidentally deleting files, sending confidential data to the wrong email, or falling for a simple scam.	A teacher emailing the whole marksheet to the wrong parent; a student forwarding an OTP "to help the bank."

Malware types in one line each: a **virus** attaches to files and spreads when you share them; a **worm** spreads by itself across a network; **spyware** secretly watches you; **ransomware** locks your files and demands money; a **Trojan** pretends to be useful software but hides harm.

Social engineering means tricking a **person**, not breaking a lock with maths. Phishing is the most common exam example.



Three high-yield threats: phishing (fake messages), DoS (server overload), identity theft (stolen CNIC or bank details).

WORKED EXAMPLE

Worked example: classify the incident.

1. Bilal downloads a "cracked" Photoshop. Next day his project files are unreadable and a note asks for Bitcoin. → **Malware** (ransomware), started by cracked software.
2. Ayesha gets an email "Your Meezan account will close in 2 hours. Click here and send OTP." The link is meezn-pk.tk, not the real bank. → **Phishing**.
3. On e-test morning the college portal spins forever; the admin graph shows 200,000 requests from one pattern of fake traffic. → **DoS**.
4. Someone opens a JazzCash wallet using Ahmed's CNIC copy from a leaked form. → **Identity theft** (after data theft).
5. The lab assistant emails the password list to the whole staff by choosing the wrong group. → **Human error**.

REMEMBER IT

Malware = bad program. Phishing = fake message. DoS = flood. Identity theft = stolen 'you'. Human error = honest mistake. Cracked software and unknown USBs are the usual door for malware in Pakistani colleges.

COMMON MISTAKE

Mistake: Calling every cyber problem a "virus", or mixing phishing with malware.

Fix: A virus is only one kind of malware. Phishing is a **trick against the user**; it may not install any file at all. DoS does not steal your password — it makes the service unusable for everyone.

اردو میں سمجھیں

سیکیورٹی خطرہ وہ عمل یا کمزوری ہے جو نظام یا ڈیٹا کو نقصان پہنچاے۔ میلویئر نقصان دہ سافٹ ویئر ہے (وائرس، ورم، سپائی ویئر، رینسم ویئر، ٹروجن)۔ فیشنگ جعلی ای میل یا ویب سائٹ سے پاس ورڈ اور او ٹی پی چھپتی ہے۔ ڈو ایس (DoS) جعلی درخواستوں کی بارش سے سائٹ کریش کر دیتا ہے۔ دیگر خطرات: ڈیٹا/شناخت کی چوری، اور انسانی غلطی۔

EXAM TIP

Expect "Define a security threat", "Explain malware / phishing / DoS with examples", and "Differentiate data theft and identity theft." List all six threats if the question says "common security threats." For DoS, write: flood of fake requests → slowdown or crash → real users blocked, and quote the online exam portal. For phishing, write social engineering + fake genuine-looking email/site + passwords/OTPs.

CHECK YOURSELF

1. Name five types of malware listed in the source.
2. What does a DoS attack do to a website?
3. How is identity theft different from data theft?

Answers: 1. Viruses, worms, spyware, ransomware, Trojans. 2. Floods it with thousands of fake requests so it slows down or crashes and real users cannot access it. 3. Data theft is stealing/copying data; identity theft is using stolen personal details (name, CNIC, bank) to pretend to be that person and commit fraud.

5.4.2 Identifying Security Threats

LEARN IT

Identifying security threats means **noticing signs** that data or systems may be at risk. Early identification helps you act before serious damage occurs.

Sign / indicator	What it likely means
Slow performance, missing files, or random pop-ups.	Malware or background unauthorised activity.
Failed login attempts or strange login locations.	Attempted unauthorised access .
Emails urging quick action, asking for an OTP, or containing bad grammar / strange links.	Classic phishing .
Unexpected deletion, copying or movement of records.	Data theft or insider misuse.
Sudden massive spike in server requests or a very slow website.	An ongoing DoS attack.

The first line of defence is also simple behaviour: avoid **unsafe downloads, cracked software, unverified websites and infected USB drives**.

WORKED EXAMPLE

Worked example: read the signs in a college computer lab in Khairpur.

1. Three PCs show pop-ups and Word files have vanished after students used a "game" USB. → Treat as **malware**. Disconnect from the network; run antivirus; stop USB use.
2. Google Classroom emails: "Your account will be deleted in 30 minutes. Confirm password here." Link spelling is go0gle. → **Phishing**. Do not click; report to the teacher.
3. The e-portal that was fine at 8:00 am is unreachable at 8:05; the technician sees a sudden spike in requests. → Suspect **DoS**.
4. Nadia's Gmail security page shows a login from another country at 4 a.m. → **Unauthorised access** attempt. Change password; turn on MFA.

REMEMBER IT

Slow/pop-ups → malware. Odd logins → unauthorised access. Urgent OTP email → phishing. Files moving → theft. Traffic spike → DoS.

COMMON MISTAKE

Mistake: Ignoring early signs ("the PC is just old") until the ransomware note appears.

Fix: One unusual login or one OTP email is enough to investigate. Early action is cheaper than recovery.

اردو میں سمجھیں

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔ سست روی، غائب فائلیں یا پاپ اپ میلویئر کی علامت ہیں؛ عجیب لاگ ان غیر مجاز رسائی کی؛ فوری او ٹی پی والی ای میل فشننگ کی؛ ریکارڈ کا اچانک منتقل ہونا ڈیٹا چوری کی؛ سرور پر درخواستوں کا دھماکہ ڈو ایس کی۔ کریک سافٹ ویئر، نامعلوم یو ایس بی اور غیر تصدیق شدہ ویب سائٹوں سے بچیں۔

CHECK YOURSELF

1. Random pop-ups and missing files most likely indicate what?
2. An email with bad grammar that demands your OTP is a sign of what?
3. Name four unsafe habits that invite these threats.

Answers: 1. Malware (or background unauthorised activity). 2. Phishing. 3. Unsafe downloads, cracked software, unverified websites, infected USB drives.

5.4.3 Apply Mitigation Techniques

LEARN IT

Mitigation techniques are actions taken to **prevent** security threats or **reduce their harmful effects** once identified. Technology helps, but it cannot stop all threats if the user is careless.

#	Technique	How it helps
1	Antivirus and firewalls	Antivirus scans and removes harmful software (for example from USB drives). A firewall monitors network traffic and blocks suspicious access from the internet.
2	Strong passwords and 2FA	Complex passwords slow brute-force guessing. 2FA adds an extra layer: an OTP on your phone as well as the password.
3	Data encryption	Converts data into unreadable code during transmission. Protects passwords and sensitive data if they are intercepted.
4	Regular backups	Copies on cloud or external drives allow quick recovery after a crash, human error or ransomware.
5	User awareness (the human factor)	Training users to avoid unknown links, suspicious attachments and password sharing. The human is often the weakest link — and the strongest defence when trained.

WORKED EXAMPLE

Worked example: match each threat to a mitigation — BISE e-test week at a college in Larkana.

Threat seen	Mitigation to apply
USB games in the lab, pop-ups starting.	Scan with antivirus ; block unknown USBs; enable the lab firewall .
Phishing SMS about “update your portal password.”	User awareness : do not click; verify on the official site typed by hand. Turn on 2FA .
Fear that intercepted Wi-Fi could show passwords.	Use sites with encryption (HTTPS / encrypted LMS); never send passwords in plain WhatsApp.
Ransomware risk to project files.	Regular 3-2-1 backups so files can be restored without paying.
Shared passwords on a sticky note on the monitor.	Strong unique passwords + 2FA; remove the note; awareness session.

REMEMBER IT

Five shields: A-F, P-2, E, B, U. Antivirus+Firewall, Passwords+2FA, Encryption, Backups, User awareness. Ransomware is why backups matter: if files are locked, a clean copy brings them back.

COMMON MISTAKE

Mistake: “If I installed antivirus I am 100% safe.”

Fix: Antivirus does not stop you from typing an OTP into a fake page. Mitigation is a **set** of techniques plus human caution.

اردو میں سمجھیں

تحفظ کی تدابیر خطرات کو روکتی یا ان کا نقصان کم کرتی ہیں: اینٹی وائرس اور فائر وال، مضبوط پاس ورڈ اور دو عنصری تصدیق (2FA)، انکریپشن، باقاعدہ بیک اپ، اور صارف کی آگاہی۔ ٹیکنالوجی تب ناکام ہو جاتی ہے جب صارف نامعلوم لنک کھولے یا پاس ورڈ بانٹے۔

CHECK YOURSELF

1. How does 2FA add a layer on top of a password?
2. Why do regular backups help after ransomware?
3. What three careless behaviours should awareness training stop?

Answers: 1. It also requires an OTP sent to your phone (a second factor). 2. You can restore a clean copy instead of losing the files (or paying). 3. Unknown links, suspicious attachments, and password sharing.

5.5 Equity and Equal Access in Digital Collaboration

LEARN IT

Collaboration tools are vital for online learning and teamwork. To use these tools fairly, all users must participate **without barriers**. You must understand the difference between **equal access** and **equity**.

Equal access

Giving everyone the **exact same** opportunity, tools and resources.

Example: providing every student with the **same login ID** to access a Zoom class.

Equity

Giving **additional, specific support** to users who need it, so they can reach the **same outcome**.

Example: turning on live **subtitles/captions** in Zoom specifically for a student who has hearing difficulties.

Equal access is the same starting kit. Equity is extra help so a person who faces a barrier can still finish the race with everyone else. Fair collaboration needs **both**: the same Zoom link for all (equal access) **and** captions, recorded lectures or a low-data app where needed (equity).

WORKED EXAMPLE

Worked example: one class, two ideas. A teacher in Thatta starts online classes.

Equal access: every student receives the same Zoom ID and password. That is fair in the sense of "same tool," but a student with hearing loss still cannot follow the spoken lecture, and a student in a village with a weak 2G signal cannot join video.

Equity: the teacher turns on captions, posts a low-size audio recording, and accepts the assignment as a Google Doc in the browser (not only through a heavy app). Now those two students can reach the **same learning outcome**.

REMEMBER IT

Equal = same tools. Equity = extra support for the same outcome. Same shoes for all vs a ramp for the wheelchair user.

COMMON MISTAKE

Mistake: "We gave everyone the same login, so the class is fair."

Fix: Same login is equal access only. If some students still cannot hear, see, afford a smartphone, or get electricity, you still need equity.

اردو میں سمجھیں

مساوی رسائی (Equal Access) سب کو ایک جیسے اوزار دیتی ہے، جیسے ایک ہی زوم لاگ ان۔ انصاف (Equity) ضرورت مند کو اضافی مدد دیتی ہے تاکہ وہی نتیجہ نکل سکے، جیسے سماعت کی مشکل والے طالب علم کے لیے زندہ ذیلی عنوانات (Captions)۔ منصفانہ تعاون کے لیے دونوں چاہییں۔

CHECK YOURSELF

1. Define equal access in one sentence.
2. Define equity in one sentence.
3. Why are Zoom captions an example of equity, not merely equal access?

Answers: 1. Giving everyone the exact same opportunity, tools and resources. 2. Giving additional, specific support so users who need it can reach the same outcome. 3. Because they are extra support for a student with hearing difficulties, not the same identical tool given to all.

5.5.1 Functions of Equity and Equal Access ★ Golden

LEARN IT

Equity and equal access are not slogans. They **do work** in a digital classroom. The source lists four functions.

Function	What it does	Pakistani example
1. Providing need-based support (equity)	Help tailored to individuals.	Audio descriptions for visually impaired students; a screen reader so a blind student in Karachi can hear a digital textbook.
2. Removing barriers	Help students in remote areas attend online classes.	Mobile-friendly, low-data apps so a student in Tharparkar or the katcha areas of Sindh can join without a fibre connection.
3. Enabling active participation	Students who miss live class can still take part.	Recorded lectures so load-shedding or an internet outage in Hyderabad does not make a student fall behind.
4. Supporting different learning needs	Offer more than one format.	Videos, text notes and audio of the same Chapter 5 topic so each student learns in the way that suits them best.

WORKED EXAMPLE

Worked example: design four functions for one class WhatsApp + Classroom group in Nawabshah.

- Need-based support:** Ali is visually impaired — share materials as selectable text (not only photos) so a screen reader can speak them; add a short audio description of each diagram.
- Remove barriers:** post a 5 MB audio summary instead of a 200 MB video for students on expensive mobile data.
- Active participation:** upload the Zoom recording within an hour; students whose lights went out at 7 pm watch it at 11 pm.
- Different learning needs:** the same lesson as (a) a 12-minute video, (b) a one-page PDF, (c) an Urdu audio note.

Discussion from the source: if a teacher **only** accepts assignments via an app that needs a **high-end smartphone**, is that equal access? **No**. It creates a **hardware barrier** for lower-income students. Web-based or light apps give better access.

REMEMBER IT

Four functions: Support, Remove, Participate, Formats. “Some Rural Pupils Fail” without them — Support (need-based), Remove barriers, Participate via recordings, Formats (video/text/audio).

COMMON MISTAKE

Mistake: Describing only captions and stopping.

Fix: Examiners want all four functions. Include remote-area low-data apps, recorded lectures for power cuts, and multiple formats — these are in the Sindh source on purpose.

اردو میں سمجھیں

انصاف اور مساوی رسائی کے چار کام ہیں: ضرورت کے مطابق مدد (جیسے نابینا طالب علم کے لیے آڈیو وضاحت)، رکاوٹیں ہٹانا (دور دراز علاقوں کے لیے کم ڈیٹا والی ایپس)، فعال شرکت ممکن بنانا (لوڈ شیڈنگ کے لیے ریکارڈ شدہ لیکچر)، اور مختلف اندازِ تعلیم کی حمایت (ویڈیو، متن، آڈیو)۔ مہنگے فون والی ایپ پر صرف اسی طرح اسائنمنٹ لینا مساوی رسائی نہیں۔

EXAM TIP

Golden question: “Explain the functions of equity and equal access in digital collaboration” (5–8 marks). Write the four headings in order, one example each, and add the hardware-barrier point (high-end app vs light/web app). Another favourite: “Differentiate equal access and equity with examples” — same login vs captions.

CHECK YOURSELF

- List the four functions of equity and equal access.
- How do recorded lectures enable active participation in Pakistan?
- Why is a high-end-smartphone-only assignment unfair?

Answers: 1. Need-based support; removing barriers; enabling active participation; supporting different learning needs. 2. Students who face power or internet outages can watch later and not fall behind. 3. It creates a hardware barrier for lower-income students; web-based or light apps give better access.

5.5.2 Collaboration Tools

LEARN IT

Digital collaboration tools are online platforms that support **communication, file sharing, learning and teamwork** across different locations.

Tool	Primary function and educational use
Google Docs	Write, edit, comment and share documents in real time. Excellent for group assignments.
Zoom	Online meetings, virtual classes, screen sharing and video-based communication.
Microsoft Teams	A unified workspace for online meetings, file sharing, class groups and structured teamwork.
Google Classroom / LMS	Learning Management Systems help teachers manage lectures, distribute assignments, conduct quizzes and grade student activities.
WhatsApp	Quick class announcements, urgent reminders and instant group communication. Must be used responsibly.
Email	Formal communication, official notices, submitting large attachments, and keeping a professional written record.

Responsibly on WhatsApp means: no leaking of classmates' marks or photos, no fake forwards, and no forcing a student who has no smartphone to miss official work that was never posted on the LMS.

WORKED EXAMPLE

Worked example: choose the right tool for each job in a Hyderabad college group project.

1. Write the report together after school, from different houses. → **Google Docs** (real-time edit and comments).
2. Sunday 7 pm teacher explanation with slides. → **Zoom** or **Teams** (video + screen share). Turn on captions (equity).
3. Collect homework, quiz and grades in one place. → **Google Classroom / LMS**.
4. "Tomorrow's test is postponed — tell everyone in the next ten minutes." → **WhatsApp** class group (then repeat on Classroom so nobody is left out).
5. Send the principal a formal complaint with a 8 MB scanned letter. → **Email** (written record + attachment).

REMEMBER IT

Docs = write together. Zoom/Teams = meet. Classroom = official class hub. WhatsApp = fast but informal. Email = formal record.

COMMON MISTAKE

Mistake: Running the whole course only on WhatsApp.

Fix: WhatsApp is for quick reminders. Assignments, marks and lectures belong on an LMS so there is access control, a record, and a path that does not depend on one person's phone.

اردو میں سمجھیں

ڈیجیٹل تعاون کے اوزار مختلف مقامات سے بات چیت، فائل شیئرنگ اور ٹیم ورک چلاتے ہیں: گوگل ڈاکس مشترکہ تحریر کے لیے، زوم اور مائیکروسافٹ ٹیمز میٹنگ کے لیے، گوگل کلاس روم یا ایل ایم ایس باقاعدہ پڑھائی کے لیے، وائس ایپ فوری اطلاع کے لیے (ذمہ داری سے)، اور ای میل رسمی ریکارڈ کے لیے۔

CHECK YOURSELF

1. Which tool is excellent for real-time group writing?
2. What four jobs does an LMS / Google Classroom help a teacher do?
3. What is email especially good for?

Answers: 1. Google Docs. 2. Manage lectures, distribute assignments, conduct quizzes, grade activities. 3. Formal communication, official notices, large attachments, and a professional written record.

Career Opportunities

LEARN IT

This unit covers neural networks, deep learning, secure collaboration, data protection and digital equity. Mastery of these ideas opens high-demand careers in **Artificial Intelligence** and **Cybersecurity** — including in Pakistan's banks, software houses, universities and public projects linked to the National AI Policy 2025.

Career	What the person does
Machine Learning Engineer	Builds and trains machine-learning and neural-network models used in face recognition, NLP and recommendation systems.
Cybersecurity Analyst	Monitors systems for threats such as malware and phishing, and applies mitigation (firewalls, 2FA) to protect organisational data.
AI Researcher	Studies and develops new techniques in artificial intelligence and deep learning, often in university labs or high-tech R&D centres.
Data Privacy Officer	Ensures the organisation's data handling complies with legal privacy principles, protecting sensitive user information and secure collaboration.
Ethical Hacker	Legally tests systems and networks for vulnerabilities (penetration testing) before malicious attackers can exploit them via DoS or malware.

WORKED EXAMPLE

Worked example: match the job.

1. "Train a model that tags Urdu news headlines." → **Machine Learning Engineer**.
2. "Watch the bank's logs for phishing domains and turn on 2FA." → **Cybersecurity Analyst**.
3. "With written permission, try to break the college portal before criminals do." → **Ethical Hacker**.

REMEMBER IT

AI careers lean on Python, maths and ethics. Cyber careers lean on threats, 2FA, firewalls and careful testing. Both need clear communication.

COMMON MISTAKE

Mistake: Confusing an ethical hacker with a criminal hacker.

Fix: An ethical hacker has **permission** and reports weaknesses so they can be fixed. Attacking without permission is a crime.

اردو میں سمجھیں

اس باب سے جڑے پیشے ہیں: مشین لرننگ انجینئر، سائبر سیکیورٹی اینالسٹ، اے آئی محقق، ڈیٹا پرائیویسی آفیسر اور اخلاقی ہیکر۔ اخلاقی ہیکر اجازت کے ساتھ کمزوریاں ڈھونڈتا ہے تاکہ مجرمانہ حملے سے پہلے انہیں ٹھیک کیا جا سکے۔

CHECK YOURSELF

1. Who legally tests a network for weaknesses before attackers do?
2. Which career focuses on compliance and protecting sensitive user information?

Answers: 1. Ethical hacker. 2. Data Privacy Officer.

Application and Impacts of Computing

17 single-page lessons · 5 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **ML** is a model (a trained brain) that learns from data in three steps: input, learning, prediction on new unseen data — without a hand-written rule for every case.
- **AI** $\supset$ **ML** $\supset$ **DL**: deep learning sits inside machine learning, which sits inside artificial intelligence.
- A **neural network** is a brain-inspired mathematical model of connected neurons in an input layer, hidden layer(s) and output layer (face unlock: camera $\rightarrow$ features $\rightarrow$ unlock).
- **Deep learning** trains neural networks with many hidden layers; early layers learn simple features, deeper layers learn complex patterns. It needs huge data and computing power.
- NN components: input layer, neurons, hidden layers, weights, bias, activation function, output layer, loss function. A neuron computes $s = \sum(x \times w) + b$, then an activation gate.
- DL components: large data, many hidden layers, automatic feature learning, learning algorithm (backpropagation), GPUs, training epochs. Pakistan's National AI Policy 2025 covers education, health, agriculture, banking and local languages.
- Applications: image and speech recognition, NLP, healthcare imaging, recommendations, autonomous vehicles, fraud detection, assistive tech.
- **Secure collaboration**: authentication (who), MFA/2FA (two or more factors), access control (what you may do), accountability, privacy, safe transmission.
- **Data protection**: encryption (plaintext $\rightarrow$ ciphertext; symmetric vs asymmetric; Caesar shift), strong passwords, 3-2-1 backup, firewall and antivirus.
- Threats: malware, phishing, unauthorised access, data/identity theft, DoS, human error. Spot them by pop-ups, odd logins, urgent OTP emails, missing files, traffic spikes.
- Mitigation: antivirus + firewall, strong passwords + 2FA, encryption, regular backups, user awareness.
- **Equal access** = same tools for all; **equity** = extra support for the same outcome. Functions: need-based support, removing barriers, recorded participation, multiple formats. Tools: Docs, Zoom, Teams, Classroom/LMS, WhatsApp, email.

Key terms

Term	Meaning	اردو
Artificial intelligence (AI)	The broad field in which machines perform tasks that normally need human thinking.	مصنوعی ذہانت
Machine learning (ML)	A model that learns patterns from data and then predicts on new data.	مشین لرننگ
Neural network (NN)	A brain-inspired mathematical model of connected neurons in layers.	نیورل نیٹ ورک
Deep learning (DL)	An ML method that trains neural networks with many hidden layers.	ڈیپ لرننگ
Weight	A number showing the strength or importance of a connection between neurons.	وزن (وزن تعلق)
Bias	An extra constant added to a neuron's sum to improve accuracy.	بائیس
Activation function	A gate that decides whether a signal is important enough to pass on.	ایکٹیویشن فنکشن
Loss function	A measure of how wrong the model's answer is compared with reality.	لاس فنکشن
Epoch	One full training pass through the dataset.	ایپوک (تربیتی چکر)
GPU	Graphics Processing Unit — specialised hardware used to train deep networks.	جی پی یو
NLP	Natural Language Processing — computers understanding and producing human language.	قدرتی زبان کی پروسیسنگ
Authentication	Confirming a user's identity before access (password, PIN, OTP, biometrics).	تصدیق شناخت

Application and Impacts of Computing

75 marks · 1 hour 15 minutes · no notes · no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 17 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. A machine-learning model works in which order?
 - a. Prediction, then data input, then learning
 - b. Data input, learning, then prediction on new data
 - c. Only prediction, with no data
 - d. Writing a separate rule for every case
2. A neural network is best described as:
 - a. a living human brain placed in a computer
 - b. a backup disk
 - c. a brain-inspired mathematical model of connected neurons
 - d. a type of firewall
3. The word "deep" in deep learning refers to:
 - a. several hidden layers through which data passes
 - b. very loud speakers
 - c. using only one neuron
 - d. storing data underground
4. In the relationship $AI \supset ML \supset DL$, the innermost set is:
 - a. AI
 - b. ML
 - c. equal access
 - d. DL
5. Weights in a neural network show the:
 - a. colour of the screen
 - b. importance or strength of connections between neurons
 - c. size of the hard disk
 - d. student's roll number
6. For $x_1 = 1$, $x_2 = 0$, $w_1 = 0.5$, $w_2 = 0.5$, $b = 0$ and threshold 0.4, the neuron output y is:
 - a. 1
 - b. 0
 - c. 0.4
 - d. 2
7. Which of the following is a component of deep learning (not of a tiny shallow network)?
 - a. Exactly one hidden layer and ten rows of data
 - b. Manual listing of every feature by a human, always
 - c. Many hidden layers, huge datasets and GPUs
 - d. A Caesar cipher
8. Unlocking a phone by matching facial features is mainly an example of:
 - a. a DoS attack
 - b. 3-2-1 backup
 - c. equal access
 - d. image recognition
9. Authentication means:
 - a. choosing who may edit a file
 - b. confirming a user's identity before allowing access
 - c. shifting letters in a cipher
 - d. flooding a server with requests
10. Password plus an OTP sent to a phone is an example of:
 - a. MFA / 2FA
 - b. a worm
 - c. a shallow neural network
 - d. equity captions

11. With a Caesar cipher of shift 3, the plaintext CAT becomes:
 - a. DBU
 - b. CAW
 - c. FDW
 - d. ZXQ
12. The 3-2-1 backup rule requires:
 - a. three passwords only
 - b. one copy on the same laptop twice
 - c. no off-site copy
 - d. three copies, two media types, one copy off-site
13. Phishing is:
 - a. a social-engineering trick using fake emails or sites to steal passwords or OTPs
 - b. a type of GPU
 - c. an activation function
 - d. a collaboration tool like Zoom
14. A Denial of Service (DoS) attack:
 - a. always steals a CNIC
 - b. floods a system with fake requests so it slows or crashes
 - c. is the same as a strong password
 - d. is a hidden layer
15. Turning on live captions in Zoom for a student with hearing difficulties is an example of:
 - a. malware
 - b. a DoS attack
 - c. equal access only, with no extra support
 - d. equity

Section B — Short questions (10 × 3 = 30)

1. Define a machine-learning model and state the three steps of how it works.
2. What is a neural network? Name its three kinds of layer and link them to face unlock.
3. Differentiate a neural network (the model) from deep learning (the method).
4. State the relationship $AI \supset ML \supset DL$ in words.
5. List eight components of a neural network. What do weights and bias do?
6. Name six components of deep learning.
7. Differentiate authentication, MFA and access control with one example each.
8. Encrypt SINDH using a Caesar cipher with shift 3. State whether this cipher is symmetric or asymmetric.
9. Define malware, phishing and DoS. Give one Pakistani-style example of each.
10. Differentiate equal access and equity. List four functions of equity and equal access.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Explain machine learning, neural networks and deep learning. Draw and label the set diagram $AI \supset ML \supset DL$. Include the face-unlock walk-through and the photo-gallery example.
2. A neuron has inputs $x_1 = 0.8$, $x_2 = 0.5$, weights $w_1 = 0.4$, $w_2 = 0.6$, bias $b = 0.1$ and threshold 0.5. (a) Compute the weighted sum s and the output y . (b) Repeat for $x_1 = 0.2$, $x_2 = 0.3$. (c) Name all eight NN components and sketch a 3-4-2 network. (d) List the six DL components and map them onto a cotton-leaf disease detector in Sindh.
3. Describe eight applications of neural networks and deep learning with examples. Then discuss the future of AI in Pakistan with reference to the National AI Policy 2025 and five sectors.
4. What is secure collaboration? Explain authentication, MFA/2FA, access control, accountability, data privacy and safe transmission, using a Google Classroom / banking-app scenario from Pakistan.
5. Define data protection. Explain encryption (symmetric vs asymmetric) with a full Caesar-cipher working of SINDH (shift 3). Describe strong passwords, the 3-2-1 backup rule, firewall and antivirus.
6. Define a security threat. Explain malware, phishing, unauthorised access, data/identity theft, DoS and human error. Give a table of identifying signs and a matching set of mitigation techniques. End with equity versus equal access and a fair choice of collaboration tools.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **5.1 Machine Learning, Neural Networks and Deep Learning.** Answers: 1. AI ($AI \supset ML \supset DL$). 2. Yes — DL is inside ML, and ML is inside AI. 3. Any of: face unlock, speech recognition, photo search, medical diagnosis, language translation.
2. **5.1.1 Machine Learning (ML).** Answers: 1. Data input, learning, prediction. 2. The trained brain of a computer program. 3. New, unseen data.
3. **5.1.2 Neural Networks (NN).** Answers: 1. Neurons or nodes. 2. Hidden layers. 3. No — it is inspired by the brain but works as a mathematical computer system.
4. **5.1.3 Deep Learning (DL).** Answers: 1. Because data passes through several hidden layers. 2. Early: simple features (edges, shapes, colours). Deeper: complex patterns (a whole face, a car model). 3. No — that is a shallow network; DL always uses many hidden layers.
5. **5.1.4 Components of Neural Networks.** Answers: 1. The importance or strength of connections between neurons. 2. $s = 1 \times 0.5 + 0 \times 0.5 + 0 = 0.5$; $0.5 \geq 0.4$ so $y = 1$. 3. The error — how much the model's answer is wrong compared with reality.
6. **5.1.5 Components of Deep Learning.** Answers: 1. It needs massive datasets (thousands or millions of records) for accuracy. 2. One training loop — one full pass through the training data. 3. GPUs (Graphics Processing Units); backpropagation.
7. **Future of AI, NN and DL in Pakistan.** Answers: 1. National AI Policy 2025. 2. Those areas often lack specialist doctors; AI can help with disease detection and medical image analysis. 3. Any three of: programming (Python), mathematics, data analysis, problem-solving, ethical use of technology.
8. **5.2 Applications of Neural Networks and Deep Learning.** Answers: 1. Speech recognition. 2. YouTube recommending more programming tutorials after a student watches computer-science videos. 3. They process real-time camera and radar data to understand roads, traffic signals, pedestrians and obstacles (for example braking for a pedestrian).
9. **5.3.1 Secure Collaboration (Authentication, MFA, Access Control).** Answers: 1. Authentication: who are you? Access control: what may you do? 2. Two-factor / multi-factor authentication (2FA/MFA). 3. Teacher shares the document as view-only with students but gives group leaders edit permission.
10. **5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall).** Answers: 1. SDN ($P \rightarrow S$, $A \rightarrow D$, $K \rightarrow N$). 2. 3 copies, 2 different media, 1 off-site copy. 3. Public key encrypts; private key decrypts.
11. **5.4.1 Common Security Threats (Malware, Phishing, DoS).** Answers: 1. Viruses, worms, spyware, ransomware, Trojans. 2. Floods it with thousands of fake requests so it slows down or crashes and real users cannot access it. 3. Data theft is stealing/copying data; identity theft is using stolen personal details (name, CNIC, bank) to pretend to be that person and commit fraud.
12. **5.4.2 Identifying Security Threats.** Answers: 1. Malware (or background unauthorised activity). 2. Phishing. 3. Unsafe downloads, cracked software, unverified websites, infected USB drives.
13. **5.4.3 Apply Mitigation Techniques.** Answers: 1. It also requires an OTP sent to your phone (a second factor). 2. You can restore a clean copy instead of losing the files (or paying). 3. Unknown links, suspicious attachments, and password sharing.
14. **5.5 Equity and Equal Access in Digital Collaboration.** Answers: 1. Giving everyone the exact same opportunity, tools and resources. 2. Giving additional, specific support so users who need it can reach the same outcome. 3. Because they are extra support for a student with hearing difficulties, not the same identical tool given to all.
15. **5.5.1 Functions of Equity and Equal Access.** Answers: 1. Need-based support; removing barriers; enabling active participation; supporting different learning needs. 2. Students who face power or internet outages can watch later and not fall behind. 3. It creates a hardware barrier for lower-income students; web-based or light apps give better access.
16. **5.5.2 Collaboration Tools.** Answers: 1. Google Docs. 2. Manage lectures, distribute assignments, conduct quizzes, grade activities. 3. Formal communication, official notices, large attachments, and a professional written record.
17. **Career Opportunities.** Answers: 1. Ethical hacker. 2. Data Privacy Officer.

Answer key

Multiple choice

1. b
2. c
3. a
4. d
5. b
6. a
7. c
8. d
9. b
10. a
11. c
12. d
13. a
14. b
15. d

Short questions — model answers

1. An ML model is the system that learns from data and gives an output — the trained brain of a program. Steps: (1) data input — massive data is given; (2) learning — it studies data and finds patterns/rules automatically; (3) prediction — after training it decides on new, unseen data.
2. A neural network is a computer model that learns patterns from data; it is inspired by the human brain but works as a mathematical system of connected neurons/nodes. Input layer receives the camera image; hidden layers study features such as eye distance and nose shape; output layer unlocks the phone if they match stored data.
3. A neural network is the model: layers, nodes and connections; it can be shallow (one or two hidden layers). Deep learning is the method of training networks that have many hidden layers, needing huge data and processing power for complex patterns.
4. Artificial intelligence is the largest set; machine learning is a part of AI; deep learning is a part of ML. All DL is ML and all ML is AI, but not all AI is ML and not all ML is DL.
5. Input layer; neurons/nodes; hidden layer(s); weights; bias; activation function; output layer; loss function. Weights show connection strength/importance. Bias is an extra constant that helps adjust the result and improve accuracy.
6. Large input data; many hidden layers; automatic feature learning; learning algorithm (e.g. backpropagation); high computing power (GPUs); training process (epochs).
7. Authentication confirms identity before access (password, PIN, OTP, biometrics — e.g. bank OTP). MFA/2FA requires two or more factor types, typically password + OTP. Access control decides what that user may do (view/edit/comment/download/share — e.g. Google Doc view-only for the class, edit for group leaders).
8. $S \rightarrow V$, $I \rightarrow L$, $N \rightarrow Q$, $D \rightarrow G$, $H \rightarrow K$ so ciphertext VLQ GK. Caesar uses the same shift key to encrypt and decrypt, so it is symmetric.
9. Malware: harmful software (viruses, worms, spyware, ransomware, Trojans) — e.g. cracked software that corrupts files. Phishing: fake genuine-looking email/SMS/site stealing passwords or OTPs — e.g. fake JazzCash OTP page. DoS: flood of fake requests that slows or crashes a service — e.g. online exam portal students cannot enter.
10. Equal access: same tools/opportunity for everyone (same Zoom login). Equity: extra specific support so people with barriers reach the same outcome (captions for hearing difficulties). Functions: need-based support; removing barriers (low-data apps for remote areas); enabling participation (recorded lectures during load-shedding); supporting different learning needs (video, text, audio).

Long questions — model answers

1. **ML**: trained-brain model; data → learn patterns → predict on unseen data; used in face unlock, speech, photo search, diagnosis, translation. **NN**: brain-inspired maths model; neurons pass information through Input, Hidden, Output; face unlock: camera (input), eye distance/nose shape (hidden), unlock (output). **DL**: ML method with multi-layer NNs; “deep” = several hidden layers; early layers: edges/shapes/colours; deeper: whole face or car model; gallery search example. **Diagram**: three nested sets labelled AI, ML, DL with AI outermost. Sentence: $DL \subset ML \subset AI$.
2. **(a)** $s = 0.8 \times 0.4 + 0.5 \times 0.6 + 0.1 = 0.32 + 0.30 + 0.1 = 0.72$; $0.72 \geq 0.5$ so $y = 1$. **(b)** $s = 0.2 \times 0.4 + 0.3 \times 0.6 + 0.1 = 0.08 + 0.18 + 0.1 = 0.36$; $0.36 < 0.5$ so $y = 0$. **(c)** Eight components as in the table; sketch three input circles, four hidden, two output, labelled layers. **(d)** Huge labelled leaf photos; many hidden layers; features learned automatically; backpropagation; GPU training; many epochs until accuracy rises — cotton pest detector for a Sindh farm.
3. Cover all eight: image recognition (face lock, cameras); speech (assistant alarm, Urdu voice); NLP (Translate, chatbots); healthcare (X-ray/CT/MRI, pneumonia); recommendations (YouTube CS videos); autonomous vehicles (brake for pedestrian); fraud (unusual bank transaction); assistive (speech-to-text). Then Policy 2025: innovation, skills, ethics, digital growth. Sectors: education (tutoring, assessments); healthcare (rural diagnosis); agriculture (crops, irrigation, yield); business/banking (fraud, chatbots); local languages (Urdu, Sindhi). Students need Python, maths, data analysis, problem-solving, ethics.
4. Secure collaboration = working together with digital tools in a safe, controlled, responsible way. Authentication: prove identity (password/PIN/OTP/biometrics); bank OTP example. MFA: two or more of know/have/are; 2FA = password + phone OTP. Access control: view/edit/comment/download/share; teacher view-only vs group-leader edit. Accountability: LMS tracks who uploaded/submitted. Privacy: protect names, marks, passwords; share only with the right people; confidentiality; legal/ethical compliance. Safe transmission: encrypted official LMS, not an open public forum. Classroom scenario: school email + MFA, role permissions, version history, no CNICs on public WhatsApp.
5. Data protection = keeping digital information safe from unauthorised access, misuse, loss, damage or accidental deletion. Encryption: plaintext → ciphertext; symmetric = same key both ways; asymmetric = public encrypt, private decrypt. Caesar shift 3: SINDH → VLQKG; decrypt by -3; wrap-around at Z. Strong password: upper, lower, number, symbol (e.g. Zk@2026#Pass) against brute force. 3-2-1: three copies, two media, one off-site (laptop + USB + Google Drive). Firewall: barrier monitoring/blocking suspicious network traffic. Antivirus: detects/removes malware on the local machine.
6. A security threat is any action, event or weakness that can harm a system or its data. Malware (cracked software); phishing (fake bank mail); unauthorised access (unlocked PC, weak/shared passwords); data theft vs identity theft (stolen CNIC used for fraud); DoS (exam portal flood); human error (wrong email, deleting files, falling for a scam). Signs table: pop-ups/missing files → malware; odd logins → unauthorised access; urgent OTP/bad grammar → phishing; records moved → theft; traffic spike → DoS. Mitigation: antivirus+firewall; strong passwords+2FA; encryption in transit; regular backups; user awareness (no unknown links, attachments, or shared passwords). Equal access vs equity; do not require a high-end phone-only app. Tools: Docs, Zoom, Teams, Classroom/LMS, WhatsApp (responsible), email (formal record).

Self-assessment: I can...

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| ☐ define ML and explain data input → learning → prediction. | ☐ draw $AI \supset ML \supset DL$ and give a system that belongs in each layer. |
| ☐ describe a neural network and the face-unlock I-H-O path. | ☐ compute a neuron's weighted sum and threshold output from numbers. |
| ☐ list eight NN components and six DL components. | ☐ give eight applications of NN/DL and Pakistan's AI policy sectors. |
| ☐ explain authentication, MFA/2FA and access control with examples. | ☐ encrypt a short word with a Caesar cipher and apply the 3-2-1 backup rule. |
| ☐ identify malware, phishing and DoS from signs and name mitigations. | ☐ distinguish equal access from equity and state four functions of equity. |
| ☐ choose a fair collaboration tool (Docs, Zoom, Teams, LMS, WhatsApp, email). | |

